



**CPT212/CPT212W – Design & Analysis of Algorithms
Semester 2, Academic Year 2025/2026**

Assignment 2

Students' declaration:

We declare that we understand what is meant by plagiarism. This assignment is all our own work and we have acknowledged any use of the published or unpublished works of other people. We hold a copy of this assignment. We can produce if the original is lost or damaged.

Student Name & Student ID	Signature	Task Distribution

Lecturer has, and may exercise, the right **NOT TO MARK** this assignment if the above declaration has NOT BEEN SIGNED and if the above declaration is FOUND TO BE FALSE, appropriate action will be taken which would lead to ZERO marks being awarded for this assignment

This **GROUP** assignment shall contribute **15%** of the overall evaluation.

ASSIGNMENT INSTRUCTIONS

The Boyer-Moore algorithm, developed by Robert S. Boyer and J Strother Moore in 1977, is a highly efficient string searching algorithm. It's a benchmark for practical string searching due to the clever approach that leverages information from the pattern itself to skip unnecessary comparisons during the search process.

TASK

Part A: Boyer-Moore Algorithm

1. Briefly explain the concept of string matching and discuss its applications in computing or real-world systems.
2. Choose a programming language of your choice, such as Python, Java, or C++, and implement the Boyer-Moore algorithm to search for a pattern within a given text. The implementation should demonstrate the correct algorithmic logic, use a clear and well-organised code structure, and include meaningful comments to explain the main steps of the algorithm. Present and analyse the output produced and discuss the strengths and weaknesses of the Boyer-Moore algorithm based on the implementation.

Part B: Variations of the Boyer-Moore Algorithm

3. Select and explain at least three variations of the Boyer-Moore algorithm, such as the Boyer-Moore-Horspool algorithm, Turbo Boyer-Moore algorithm, Sunday/Quick Search algorithm, or other relevant Boyer-Moore-based variations. For each selected variation, briefly explain its main idea and describe the key differences compared with the original Boyer-Moore algorithm.
4. Compare and contrast the original Boyer-Moore algorithm with the three selected variations in terms of their time complexity, space complexity, strengths, weaknesses, and overall suitability or performance considerations. Include relevant pseudocode to illustrate and support the discussion of computational complexity.

Part C: Comparison with the KMP Algorithm

5. Briefly explain the Knuth-Morris-Pratt (KMP) algorithm as another string-matching algorithm. The explanation should describe its main working principle and how it avoids unnecessary character comparisons.
6. Compare and contrast the Boyer-Moore algorithm with the KMP algorithm in terms of their time complexity, space complexity, strengths, and weaknesses. Include relevant pseudocode to illustrate and support the complexity analysis.
7. Discuss scenarios in which one algorithm may be more preferable than the other. Justify the discussion using relevant characteristics of the algorithms, such as text size, pattern length, preprocessing needs, expected performance, or application context.

Document the tasks in a report.

REQUIREMENTS:

Online Submission

- Upload the assignment to eLearning. Only one submission per group is required. The submission due date is 14 June 2026 (11.59 PM).

Report Format

- The FORMAT of the assignment should be Font Name/Size: Times New Roman, 12pt; Line spacing of 1.5; Justified (Alignment); Page Number should be typed. The Table of Contents should be included.

Important Note

- 100% Plagiarism is not tolerated. Please rephrase.
- Provide references (or any PROMPT from any AI tools that you used). If the rephrasing and references are not provided, a deduction of the marks 10% for each part.
- If the report appears suspicious, all group members might be called to present the assignment.
- Marks will be deducted for late submissions accordingly (10%)

Group: (Minimum: 2 persons; Maximum: 3 persons). Everyone in the group must contribute. Provide which person working for which part of the assignment.

Contribution: 15% of coursework.

Due date: 14 June 2026 (11.59pm)

Grading Rubric

Part	Criteria	CLO	Poor (1–2 marks)	Average (3–4 marks)	Good (5–7 marks)	Excellent (8–10 marks)	Weight
Part A	1. Explanation of string matching and applications	CLO2	Incomplete, inaccurate, or unclear explanation of string matching. Applications are missing or irrelevant.	Provides a basic explanation of string matching with limited or partially relevant applications.	Clearly explains string matching and provides relevant application examples.	Provides a comprehensive and precise explanation of string matching with multiple well-chosen real-world applications.	8
Part A	2. Implementation of Boyer-Moore algorithm: correct logic, code structure, comments, output analysis, and discussion of strengths/weaknesses	CLO2	Implementation contains major logic errors or is incomplete. Code structure, comments, output analysis, and discussion of strengths/weaknesses are weak or missing.	Implementation is partially correct with some logical issues. Code structure and comments are limited; output analysis and discussion are superficial.	Implementation is mostly correct and organized. Comments are adequate, and output analysis with discussion of strengths/weaknesses is reasonably explained.	Implementation is fully correct, well-structured, and clearly commented. Output analysis is accurate, with insightful discussion of strengths and weaknesses.	25
Part B	3. Explanation of at least three Boyer-Moore variations, e.g., Boyer-Moore-Horspool, Turbo Boyer-Moore, and Sunday/Quick Search algorithm	CLO2	Fewer than three variations are explained, or explanations are incorrect, vague, or missing.	Identifies at least three variations, but explanations are basic, incomplete, or lack clarity.	Clearly explains at least three variations and describes their main operational differences from the original Boyer-Moore algorithm.	Provides detailed, accurate, and well-organized explanations of at least three variations, supported by suitable examples, illustrations, or comparison points.	13
Part B	4. Comparison of the original Boyer-Moore algorithm with at least three variations: time complexity, space complexity, strengths, weaknesses, conclusion, and pseudocode to illustrate complexity	CLO3	Comparison is incomplete, inaccurate, or covers fewer than three variations. Time/space complexity, strengths/weaknesses, conclusion, and pseudocode are missing or poorly presented.	Provides a basic comparison involving at least three variations, but analysis of time/space complexity, strengths, weaknesses, and conclusion is limited. Pseudocode is included but does not clearly support	Presents a mostly accurate comparison of the original Boyer-Moore algorithm with at least three variations, with reasonable discussion of time/space complexity, strengths, weaknesses, and conclusion. Relevant pseudocode is provided and partially supports the complexity analysis.	Delivers a thorough and insightful comparison of the original Boyer-Moore algorithm with at least three variations, with accurate time and space complexity analysis, well-justified strengths and weaknesses, and a strong conclusion.	20

				the complexity discussion.		Appropriate pseudocode clearly illustrates and supports the complexity discussion.	
Part C	5. Explanation of KMP algorithm	CLO2	Explanation is incorrect, confusing, or incomplete.	Provides a partial explanation of the KMP algorithm with limited clarity.	Clearly explains the KMP algorithm with mostly accurate details.	Provides a detailed and accurate explanation of KMP, including its main mechanism, supported by a relevant example, illustration, or pseudocode.	14
Part C	6. Comparison with KMP: time complexity, space complexity, strengths, weaknesses, summary, and pseudocode to illustrate complexity	CLO3	Comparison is incomplete, inaccurate, or lacks meaningful reasoning. Time/space complexity, strengths/weaknesses, summary, and pseudocode are missing or poorly presented.	Provides a basic comparison with minimal discussion of time/space complexity, strengths, weaknesses, and summary. Pseudocode is included but does not clearly support the complexity discussion.	Presents an accurate comparison with reasonable discussion of time/space complexity, strengths, weaknesses, and summary. Relevant pseudocode is provided and partially supports the complexity analysis.	Provides a complete, clear, and insightful comparison with sound time and space complexity analysis, justified strengths and weaknesses, and a well-supported summary. Appropriate pseudocode clearly illustrates and supports the complexity discussion.	13
Part C	7. Scenarios where one algorithm is preferable	CLO2	Scenarios are missing, inaccurate, or poorly justified.	Identifies one or a few scenarios with weak or general justification.	Provides suitable scenarios with some explanation of algorithm suitability.	Presents multiple well-justified scenarios that clearly explain when one algorithm is preferable over another.	7
	Total						100